JANMITRA

Civic Issue Reporting & Management System

Nagpur Municipal Corporation | 4 Categories | 4 Wards | 4 Departments

Team ResolveIT

Department-Wise Dashboard Specification

Detailed specification for the department dashboard feature

CITIZEN PORTAL	Report Issue | Track Status | Verify Resolution
BACKEND (Django)	REST API | Auth | Business Logic | AI Module
AI VALIDATION	Fake Detection (EXIF+ELA+HuggingFace) | YOLO Relevance Check
DATABASE	PostgreSQL + PostGIS | 4 Categories | 4 Wards | 4 Depts
DEPT. DASHBOARD	PWD | Solid Waste | Water Works | Electrical

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1. Overview

The Department-Wise Dashboard is a dedicated analytics and management interface for each of the 4 NMC departments. It gives department officers and the admin real-time visibility into complaint volume, worker performance, ward distribution, resolution speed, and open issues. This is one of JANMITRA's key value propositions for the NMC.

WARN	Data Isolation Rule: A Dept. Officer user can ONLY see data for their own department. This is enforced at the API level by always filtering queries by request.user.department_id. Admin users can view all departments and switch between them.

2. The Four Department Dashboards

Department	Dashboard URL Path	Handles Category
Public Works Department (PWD)	/officer/dashboard	Roads & Infrastructure
Solid Waste Management Dept. (SWM)	/officer/dashboard	Garbage & Waste Management
Public Health Engineering / Water Works	/officer/dashboard	Water & Drainage
Electrical Department	/officer/dashboard	Street Lighting & Electrical

All departments use the same dashboard URL (/officer/dashboard). The system auto-loads data for the logged-in officer's department. Admin uses /admin/dashboard to see all.

3. Dashboard Widgets -- Detailed Specification

Widget 1: KPI Summary Cards (Top Row)

Card	Metric	Colour	API Source
Total Complaints	All complaints in this dept. in selected date range	Blue	GET /api/v1/dashboard/kpi/
Pending	Status = Routed or Validated (not yet assigned)	Amber	GET /api/v1/dashboard/kpi/
In-Progress	Status = Assigned or In-Progress	Purple	GET /api/v1/dashboard/kpi/
Resolved	Status = Closed	Green	GET /api/v1/dashboard/kpi/
Rejected / Invalid	Status = Invalid or Rejected-Fake	Red	GET /api/v1/dashboard/kpi/

Widget 2: Average Resolution Time

Metric	Average number of days from complaint submission to closure (Closed status)
Display	Single large number card: e.g. '3.2 days'. Secondary: 'Target: under 5 days'
Calculation	avg(resolved_at - submitted_at) for all Closed complaints in date range for this dept.
Filters	Affected by date range filter and ward filter

Widget 3: Ward-Wise Complaint Count (Bar Chart)

Chart Type	Vertical bar chart (Recharts BarChart)
X Axis	Ward 1, Ward 2, Ward 3, Ward 4
Y Axis	Number of complaints
Colour	Bars grouped by status: Pending (amber), In-Progress (purple), Resolved (green), Rejected (red)
API	GET /api/v1/dashboard/ward-stats/ returns: [{ward:1, pending:5, in_progress:3, resolved:12,

	rejected:1}, ...]
Click action	Clicking a ward bar filters the High-Priority list and Worker Workload below to show only that ward

Widget 4: Complaint Severity Distribution (Pie Chart)

Chart Type	Donut chart (Recharts PieChart)
Segments	Critical (red), High (orange), Medium (amber), Low (grey)
Data	Count of complaints by priority_level for current dept + date range
Legend	Below chart: colour + label + count + percentage
API	GET /api/v1/dashboard/kpi/ includes severity_distribution: [{level, count, pct}]

Widget 5: Complaint Trend Over Time (Line Chart)

Chart Type	Line chart (Recharts LineChart)
X Axis	Date (day / week / month -- toggle button)
Y Axis	Number of complaints submitted
Lines	One line for Total, one for Resolved (to show resolution rate trend)
API	GET /api/v1/dashboard/trends/?groupby=day|week|month returns [{date, submitted, resolved}]
Default	Last 30 days, grouped by day

Widget 6: Worker Workload Table

Column	Content
Worker Name	Full name of the worker
Ward	Worker's assigned ward (Ward 1-4)
Active Tasks	Number of complaints currently In-Progress or Assigned to this worker
Completed This Week	Complaints closed in last 7 days by this worker
Avg. Resolution Time	Average days to close a complaint for this worker
Status	Active / On Leave / Inactive

Filter: Ward dropdown filters table to workers in that ward only. Useful when officer manages workers across multiple wards.

Widget 7: High-Priority Open Complaints

Column	Content
Complaint ID	Unique ID (clickable -- opens complaint detail)
Category	One of 4 categories with CategoryBadge
Ward	Ward 1-4 with WardBadge
Priority	Critical or High with PriorityBadge
Status	Current status with StatusBadge
Days Open	Days since submission. Red if > 5 days.
Assigned To	Worker name or 'Unassigned' (red text if unassigned)

List shows only Critical and High priority complaints that are NOT yet Closed. Sorted by days open (oldest first).

Widget 8: Pending AI Validation Queue

Purpose

Complaints where YOLO AI confidence was below 0.5. Officer must manually validate or reject.

Display

	Card per complaint: complaint photo thumbnail, AI confidence bar (red if < 30%, amber 30-50%), description, category, ward
Actions	'Validate and Route' green button (sets status=Validated, triggers routing). 'Reject as Invalid' red button (requires reason text).
API	GET /api/v1/dashboard/pending-validation/ filtered by officer's dept. PATCH /api/v1/dashboard/complaints/{id}/validate/

4. Dashboard Filters

INFO All filters affect all widgets on the dashboard simultaneously. Filter state persists while on the dashboard page.

Filter	Options	Affects
Date Range	Last 7 days / Last 30 days / Last 90 days / Custom fro	All widgets
Ward	All Wards / Ward 1 / Ward 2 / Ward 3 / Ward 4	All widgets
Status	All / Pending / In-Progress / Resolved / Rejected	KPI, High-Priority, Workload
Severity	All / Critical / High / Medium / Low	KPI, High-Priority, Pie chart

5. Admin Cross-Department View

The admin can view the dashboard for all 4 departments. Admin dashboard has an additional top-level widget:

Admin Widget	Content
Dept. Comparison Cards	4 cards side by side: one per dept. Each shows Total, Pending, Resolved, Avg Resolution Time
Cross-Dept Bar Chart	Category on X axis (4 bars), Total complaints on Y. Compare complaint load across departments.
Dept. Switch Dropdown	At top of page: 'Viewing: All Departments'. Dropdown to switch to single dept. view.
Worker Reg. Button	Quick link to Worker Management to register new workers for any dept+ward combination.

6. Dashboard API Design

Endpoint	Method	Params	Response
/api/v1/dashboard/kpi/	GET	dept_id, ward_id, date_from, date_to,	total, pending, in_progress, resolved, rejected, avg
/api/v1/dashboard/ward-stats/	GET	dept_id, date_from, date_to	per_ward: {ward_id, ward_name, pending, in_prog
/api/v1/dashboard/trends/	GET	dept_id, groupby (day/week/month), d	date_series: [{date, submitted, resolved}]
/api/v1/dashboard/workers/	GET	dept_id, ward_id	workers: [{id, name, ward, active, completed_weel
/api/v1/dashboard/high-priority/	GET	dept_id, ward_id	complaints: [{id, category, ward, priority, status, da
/api/v1/dashboard/pending-validation/	GET	dept_id	complaints: [{id, category, ward, ai_confidence, im

All endpoints: IsDeptOfficer permission always forces dept_id = request.user.department_id. IsAdmin can pass any dept_id or omit for all.